

Case Swarm

Technical Architecture

Capability view: protocols, APIs, credentials, data stores.
Includes visual diagrams (PNG) and boxed text architecture / data-flow diagrams.

Architecture - system context

Capability boxes only. Two outbound Microsoft auth surfaces (Graph vs Bot Framework).

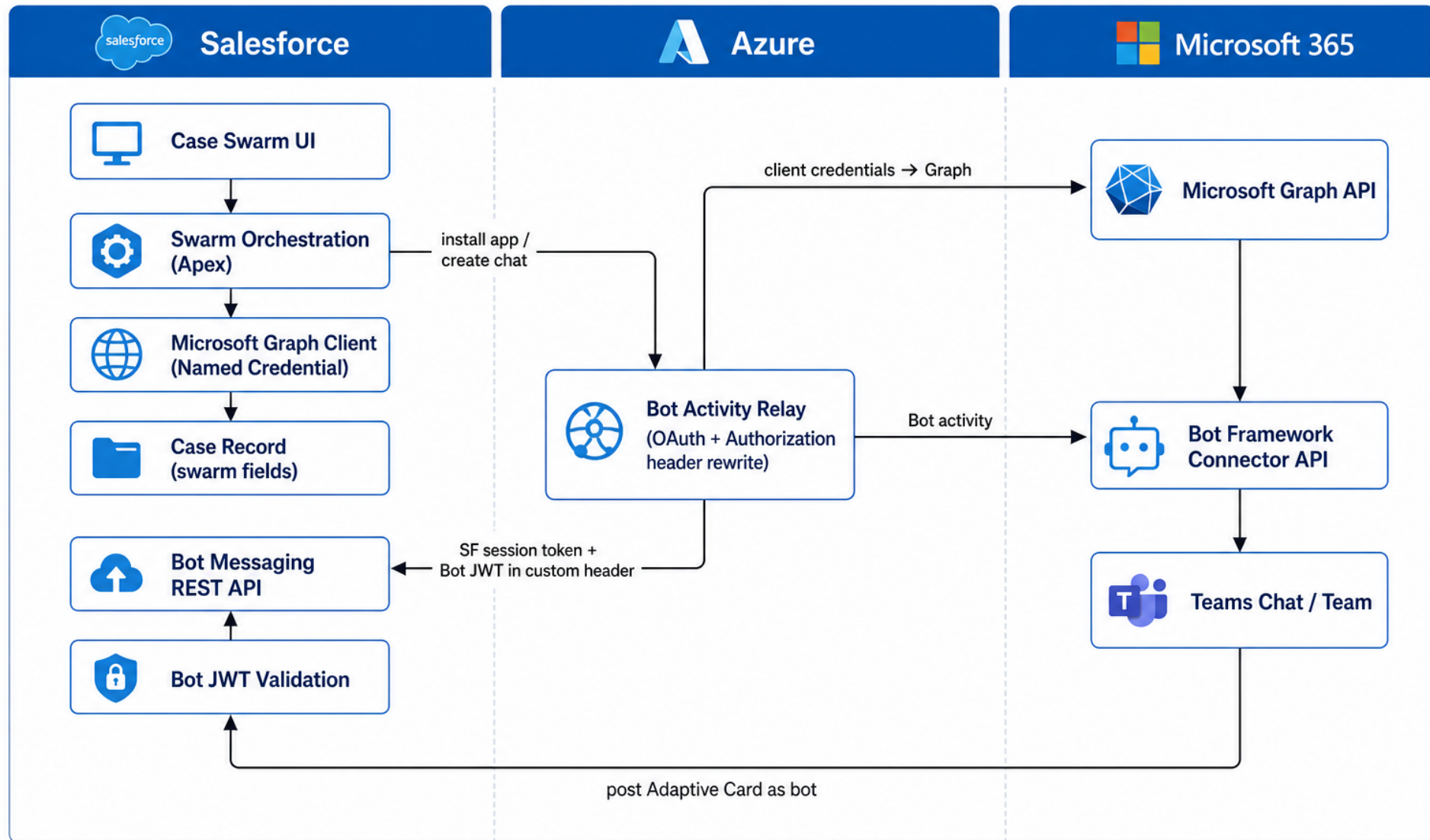


Auth surfaces (do not conflate)

Integration	Protocol	Purpose
Graph Named Credential	Client credentials	Create chat/group/team, members, list, install app
Bot Framework Named Credential	Client credentials	Post Adaptive Cards and live chat text as bot
Bot Activity Relay -> Salesforce	Connected App OAuth	Inbound bot activities; Bot JWT in custom header

Graph app-only cannot post interactive live chat (401). Outbound chat/cards use Bot Framework. Salesforce rejects Bot JWT in Authorization; relay moves it to a custom header for Apex validation.

1a. Architecture diagram - swarm provision + Adaptive Card



1b. Data flow (boxed) - swarm provision + Adaptive Card

No chat transcript object written in this flow. Chat path shown; Team path uses group + async enable Team.

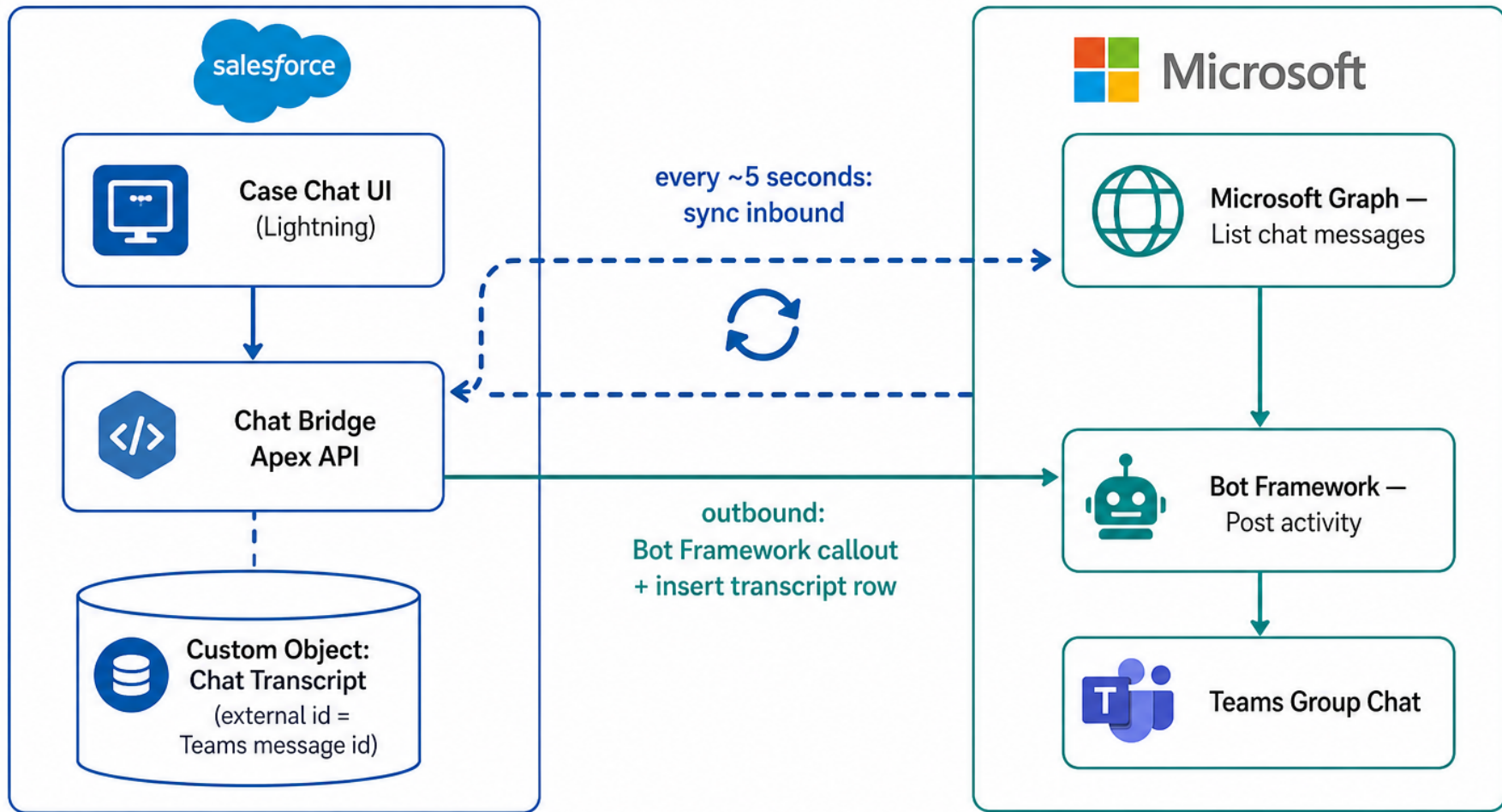
```
[Case Swarm UI]
|
v start swarm (Chat) + members
[Swarm Orchestration]
|
+-----> [NC Graph] ----> POST /chats ----> [Teams Chat]
|                                     ^
v                                     |
[Case: Active, type=Chat, chat id, URL] |
|                                     |
v async                             |
[Bot app install via Graph] -----+
|
v bot-added activity
[Teams] --> [Bot Framework] --> [Bot Activity Relay]
|
v SF token + Bot JWT (custom header)
[Bot Messaging REST API]
| validate JWT
| store Bot service URL on Case
v
[NC Bot Framework] --> post Adaptive Card --> [Teams]
```

Why relay: Bot Framework always sends Authorization Bearer Bot-JWT; Salesforce Apex REST treats Authorization as a Salesforce session and would 401 before Apex. Relay swaps tokens.

2a. Architecture diagram - Apex poll chat sync

Azure chat bridge feature flag OFF.

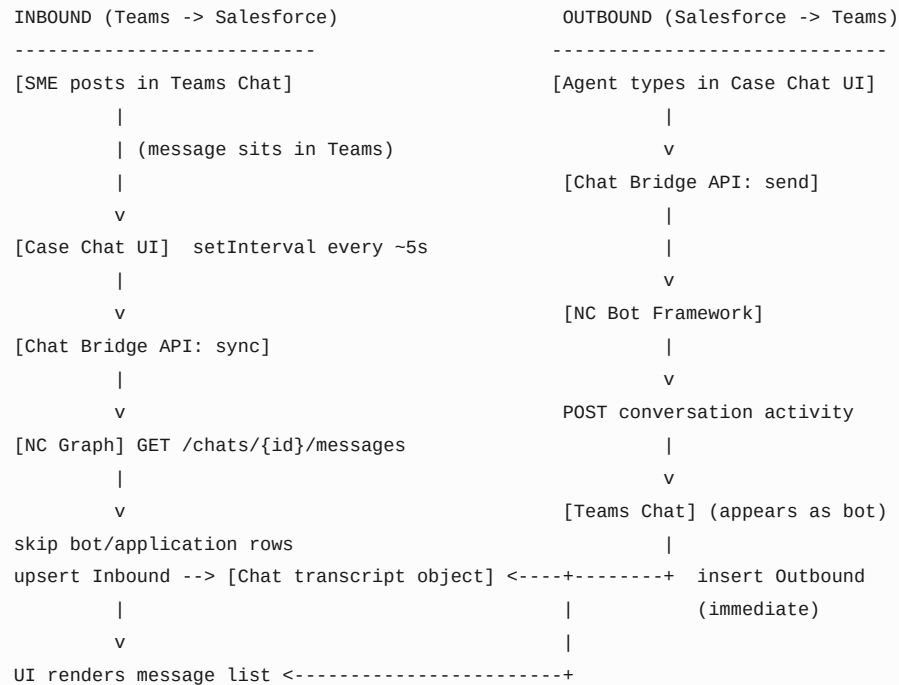
Case Swarm Chat — Apex Poll Synchronization



Feature flag off: Salesforce polls Graph • 1 callout per poll tick • 1 callout per send

2b. Data flow (boxed) - Apex poll chat sync

Condition: Use_Azure_Chat_Bridge = false (or Azure session fallback). No Graph webhook - Salesforce PULLS messages on a timer.



Exact poll call chain (code path)

```
Case Chat UI: startPolling()           // setInterval 5000 ms
    -> syncQuietly()                     // if NOT Azure bridge
    -> fetchApexSync()
    -> Chat Bridge API: syncMessages(caseId)
    -> upsertFromGraph(caseId, includeBot=false)
    -> Graph client: listChatMessages(chatId, 50)
    -> Http.send GET callout:MS_Graph/.../chats/{id}/messages
    -> upsert Chat transcript custom object

Azure bridge ON: same timer, but syncQuietly -> fetchAzureHistory()
(browser -> Azure -> Graph). Does NOT call syncMessages each tick.
```


Chat transcript persistence (Apex poll mode)

Direction	Mechanism	When written	Notes
Outbound	Bot Framework callout + DML insert	Immediate	External id = bot activity id
Inbound	Graph list + DML upsert	Next poll ~5s	External id = Teams message id
Bot Graph echo	Filtered on sync	Not stored again	Avoid duplicate Outbound
Card field edit	Bot Messaging REST -> Case DML	Case fields only	No transcript row

Runtime cost (architect view)

Operation	Salesforce impact
Chat panel open ~45 min	~540 Apex HTTP callouts to Graph (not Daily REST API)
Each agent send	1 Bot Framework callout + 1 DML
Bot install / card invoke	Inbound Daily REST API via relay

Webhook?

No Microsoft Graph change-notification webhook for chat messages. Inbound chat is always pull (Apex poll or Azure history poll). Bot Framework POSTs to the Azure relay only for bot install and Adaptive Card invoke (bot messaging endpoint, not a Graph chat webhook).

Capability map

Case Swarm UI / Case Chat UI (Lightning) | Swarm Orchestration + async provision (Apex) | Chat Bridge API sync/send (Apex) | Bot Messaging REST + JWT validation | Bot Activity Relay (Azure Functions) | Named Credentials: Graph + Bot Framework | Transcript custom object (external id = Teams message id) | Identity: Azure AD email/UPN on User